2020 - A

1. $\lim_{z\to 0} \frac{\bar{z}^2}{z} = 0$

证明：$z = re^{i\theta}$ $\bar{z} = re^{-i\theta}$

左式 $= \lim_{r\to 0} \frac{r^2 e^{i(-2\theta)}}{re^{i\theta}} = \lim_{r\to 0} r e^{i(-3\theta)}$ $\because |e^{i(-3\theta)}| = 1$

$\therefore \lim_{z\to 0} \frac{\bar{z}^2}{z} = 0$

2. 在映射 $w = z^2$ 下，$x^2 - y^2 = 3$ 象曲线

$z = x + iy$

$w = z^2 = x^2 - y^2 + i\,2xy$

$\begin{cases} u = x^2 - y^2 \\ v = 2xy \end{cases}$

即 $u = 3$ $(v \in R)$

处理z的共轭

3. $\int_{|z|=2} \frac{e^z}{(z-1)^2(1-\bar{z})} dz$ 积分路径正向

$= \int_{|z|=2} \frac{z e^z}{(z-1)^2 (z-4)} dz$ // 有 $\bar{z}$ 先消去. $z \cdot \bar{z} = re^{i\theta} \cdot re^{-i\theta} = r^2$

$= \frac{2\pi i}{1!} \left(\frac{ze^z}{z-4}\right)' \Big|_{z=1} = 2\pi i \operatorname{Res}[f(z), 1] = 2\pi i \frac{1}{1!} \lim_{z\to 1} \left(\frac{ze^z}{z-4}\right)'$

$= 2\pi i \frac{(ze^z + e^z)(z-4) - ze^z}{(z-4)^2} \Big|_{z=1} = 2\pi i \frac{-7e}{9} = \frac{-14\pi e i}{9}$

高阶导数定理

由柯西积分定理：

① $\int_C \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0)$ （f(z)解析，C内含 z_0 这个奇点）

两端对 z_0 求一阶导：

$$\int_C \frac{f(z)}{(z-z_0)^2} dz = 2\pi i f'(z_0)$$

② 如果是求n阶导：

$$n! \int_C \frac{f(z)}{(z-z_0)^{n+1}} = 2\pi i f^{(n)}(z_0)$$

关于负号：

对 $(z-z_0)^{-n}$ 求导：

$(-n)(z-z_0)^{-n-1} \cdot (-1)$

$= n(z-z_0)^{-n-1}$

所以最后没负号

$-z_0$ 的负号

一样的

留数定理

$$\oint_C f(z)\, dz = 2\pi i \sum_{k=1}^{n} \mathrm{Res}[f(z), z_k]$$

$$\mathrm{Res}[f(z), z_0] = \frac{1}{(n-1)!} \lim_{z \to z_0} \frac{d^{n-1}}{dz^{n-1}} [(z-z_0)^n f(z)]$$

4. $\oint_C \frac{e^{\pi+z}}{z(1-z)^2}dz$ C为不过0和1的正向简单闭曲线

① 当曲线C所围区域内不包含 $z_0=0$ 和 $z_1=1$ 时

$$I=\oint_C \frac{e^{\pi+z}}{z(1-z)^2}dz=0$$ (处处解析)

$ie^{\pi}=$

② 当曲线C所围区域内不包含 $z_1=1$，包含 $z_0=0$ 时

$$I=\oint_C \frac{\frac{e^{\pi+z}}{(1-z)^2}}{z}dz = 2\pi i \left.\frac{e^{\pi+z}}{(1-z)^2}\right|_{z=0} = 2\pi i e^{\pi}$$

③ 当曲线C所围区域内不包含 $z_0=0$，包含 $z_1=1$ 时

$$I= 2\pi i \operatorname{Res}\left[\frac{e^{\pi+z}}{z(1-z)^2},1\right]=2\pi i \frac{1}{1!}\lim_{z\to 1}\frac{d\left(\frac{e^{\pi+z}}{z}\right)}{dz}$$

$$=2\pi i\, e^{\pi}\left.\frac{ze^{z}-e^{z}}{z^2}\right|_{z=1} = 2\pi i\cdot 0=0$$

④ 当曲线C所围区域内既包含 $z_0=0$ 也包含 $z_1=1$ 时

$$I=2\pi i\left(\operatorname{Res}\left[\frac{e^{\pi+z}}{z(1-z)^2},0\right]+\operatorname{Res}\left[\frac{e^{\pi+z}}{z(1-z)^2},1\right]\right)=2\pi i e^{\pi}$$

5. $$\oint_{|z|=3} \frac{z^5 \sin z}{(z^4+1)^3}\, dz$$

$$= \int_{|\xi|=\frac{1}{3}} \frac{\frac{1}{\xi^5}\sin\frac{1}{\xi}}{\left(\frac{1}{\xi^4}+1\right)^3}\, \frac{1}{\xi^2}\, d\xi$$

$$= \int_{|\xi|=\frac{1}{3}} \frac{\xi^5 \sin\frac{1}{\xi}}{(1+\xi^4)^3}\, d\xi$$

有极点 $\xi=0$

$$= 2\pi i \, \mathrm{Res}[f(\xi), 0]$$

$$= 2\pi i \cdot 0$$

$$= 0$$

$\frac{\xi^5 \sin\frac{1}{\xi}}{(1+\xi^4)^3}$ 是偶函数，$C_{-1}=0$

可以发现在 $|z|=3$ 内有太多奇点

▨ 积分换元 （留数定理 2）

$$\int_{|z|=R} f(z)\, dz = \int_{|\xi|=\frac{1}{R}} f\left(\frac{1}{\xi}\right) \frac{1}{\xi^2}\, d\xi$$

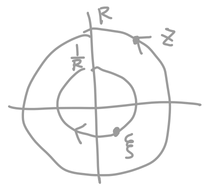
负号去哪儿了?

$\xi = \frac{1}{z}$

$d\xi = -\frac{1}{z^2}\, dz$

$z = re^{i\theta}$

$\xi = \frac{1}{r}e^{i\theta}$



z和ξ的积分方向不同，所以还会产生一个负号、

处理sin1/z

算它相关的留数

代入 sin z 展开.

P138 – P139

$$\oint_{|z|=2} \frac{\sin\frac{1}{z}}{1-z}\, dz$$

求解析函数原函数

6、解析函数 $f(z)=u(x,y)+iv(x,y)$

$$u+v=(x-y)(x^2+4xy+y^2)-2(x+y)$$
$$= x^3+4x^2y+xy^2-x^2y-4xy^2-y^3-2x-2y$$
$$= x^3-y^3+3x^2y-3xy^2-2x-2y$$

$$u'_x+v'_x= 3x^2+6xy-3y^2-2$$
$$u'_y+v'_y= -3y^2+3x^2-6xy-2$$

设 $f'(z)=a+ib$

$$\begin{cases} u'_x=v'_y=a \\ -u'_y=v'_x=b \end{cases}$$

$$\begin{cases} a+b= 3x^2+6xy-3y^2-2 \\ -b+a= -3y^2+3x^2-6xy-2 \end{cases} \Rightarrow \begin{cases} a= 3x^2-3y^2-2 \\ b= 6xy \end{cases}$$

★找到这个！ $f'(z)= 3x^2-3y^2-2 + i\,6xy$

$= 3z^2-2$

// 代入 $y=0$ 得 $f'(z|_{y=0})=3x^2-2$

$\therefore f'(z)=3z^2-2$

$f(z)= z^3-2z+C$ $\quad$ 又∵ $u+v=(x-y)(x^2+4xy+y^2)-2(x+y)$

$\therefore C=0 \quad f(z)=z^3-2z$

对f(z)求导-推导过程

$f(z)=w(x,y)=u(x,y)+i\,v(x,y)$

$dw=f'(z)\,dz$

$du+idv = (a+ib)(dx+idy)$

$du+idv = (adx-bdy)+i(ady+bdx)$

$du=adx-bdy \Rightarrow u'_x=a \quad u'_y=-b$

$dv=ady+bdx \Rightarrow v'_x=b \quad v'_y=a$

7. $C:|z|=1$ $\quad \int_{|z|=1} \frac{1}{z+2}\,dz$ 并证 $\int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta = 0$

$\frac{1}{z+2}$ 有一奇点 $z=-2$ 在C外、

$\therefore$ $\frac{1}{z+2}$ 在C及C内解析 $\quad \therefore \int_{|z|=1} \frac{1}{z+2}\,dz = 0$

$z = e^{i\theta} = \cos\theta + i\sin\theta$

$dz = (-\sin\theta + i\cos\theta)\,d\theta$

$$\int_{|z|=1} \frac{1}{z+2}\,dz = \int_0^{2\pi} \frac{-\sin\theta + i\cos\theta}{(\cos\theta+2) + i\sin\theta}\,d\theta$$

$$= \int_0^{2\pi} \frac{(-\sin\theta + i\cos\theta)(\cos\theta+2-i\sin\theta)}{(\cos\theta+2+i\sin\theta)(\cos\theta+2-i\sin\theta)}\,d\theta$$

$$= \int_0^{2\pi} \frac{-\cancel{\sin\theta\cos\theta} - 2\sin\theta + i\sin^2\theta + i\cos^2\theta + 2i\cos\theta + \cancel{\sin\theta\cos\theta}}{\cos^2\theta + 4\cos\theta + 4 + \sin^2\theta}\,d\theta$$

$$= \int_0^{2\pi} \frac{i(1+2\cos\theta) - 2\sin\theta}{5+4\cos\theta}\,d\theta = 0$$

实部、虚部均为0

$$\int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta$$

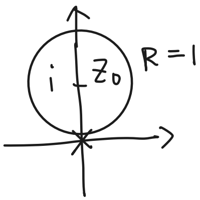
$$= \int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta + \int_{\pi}^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta$$ 设 $\varphi = 2\pi-\theta$

$$= \int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta + \int_{\pi}^{0} \frac{1+2\cos(2\pi-\theta)}{5+4\cos(2\pi-\theta)}(-1)\,d\theta$$

$$= 2\int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta = 0$$

$$\therefore \int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\,d\theta = 0$$

8. $f(z)=\dfrac{(z-i)^3}{z^8}$ 　$z_0=i$ 处泰勒展开. 收敛半径.



① 判断泰勒展开的区域.

② $f(z)=(z-i)^3\,(i+z-i)^{-8}$

$=(z-i)^3\left(1+\dfrac{z-i}{i}\right)^{-8}$ 　　$i^{-8}=\dfrac{1}{i^8}=1$

$$(1+z)^{\alpha}=1+\sum_{n=1}^{\infty}\frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}z^n$$

代入 $z=\dfrac{z-i}{i}$

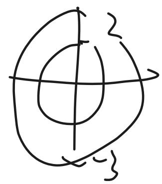
一共有 n 项

$$f(z)=(z-i)^3\left[1+\sum_{n=1}^{\infty}\frac{(-8)(-9)\cdots(-7-n)}{n!}\frac{(z-i)^n}{i^n}\right]$$

$$=(z-i)^3\left[1+\sum_{n=1}^{\infty}\frac{i^n\,(n+7)!}{n!\;7!}(z-i)^n\right]$$

$$=(z-i)^3+\sum_{n=1}^{\infty}\frac{i^n(n+7)!}{7!\,n!}(z-i)^{n+3}$$ 　$(R=1)$

9. $f(z)=\dfrac{z-1}{(z-2)(z+3)}$ 　$2<|z|<3$ 和 $|z|>3$



$f(z)=\dfrac{A}{z-2}+\dfrac{B}{z+3}$

$A=\left.\dfrac{z-1}{z+3}\right|_{z=2}=\dfrac{1}{5}$

$B=\left.\dfrac{z-1}{z-2}\right|_{z=-3}=\dfrac{4}{5}$ 　　$(1-z)^{-1}=\sum\limits_{n=0}^{\infty}z^n$ 　$|z|<1$

$f(z)=\dfrac{1}{5}\left(\dfrac{1}{z-2}+\dfrac{4}{z+3}\right)$

① $2<|z|<3$

$f(z)=\dfrac{1}{5}\dfrac{\frac{1}{z}}{1-\frac{2}{z}}+\dfrac{1}{5}\dfrac{\frac{4}{3}}{1+\frac{z}{3}}$ 　模小于1 才能展

$=\dfrac{z^{-1}}{5}\sum\limits_{n=0}^{\infty}\left(\dfrac{2}{z}\right)^n+\dfrac{4}{15}\sum\limits_{n=0}^{\infty}\left(-\dfrac{z}{3}\right)^n$

$=\dfrac{1}{5}\sum\limits_{n=0}^{\infty}2^n z^{-n-1}+\dfrac{4}{15}\sum\limits_{n=0}^{\infty}(-3)^{-n}z^{-n}$

② $|z|>3$

$f(z)=\dfrac{1}{5}\dfrac{\frac{1}{z}}{1-\frac{2}{z}}+\dfrac{1}{5}\dfrac{\frac{4}{z}}{1+\frac{3}{z}}$

$=\dfrac{z^{-1}}{5}\sum\limits_{n=0}^{\infty}\left(\dfrac{2}{z}\right)^n+\dfrac{4z^{-1}}{5}\sum\limits_{n=0}^{\infty}\left(-\dfrac{3}{z}\right)^n$

$=\dfrac{1}{5}\sum\limits_{n=0}^{\infty}2^n z^{-n-1}+\dfrac{4}{5}\sum\limits_{n=0}^{\infty}(-3)^n z^{-n-1}$

10. $$\int_{|z|=2} \frac{dz}{(z-4)(z^5-1)}$$

$$= \int_{|\xi|=\frac{1}{2}} \frac{1}{(\xi^{-1}-4)(\xi^{-5}-1)} \xi^{-2} \, d\xi$$

$$= \int_{|\xi|=\frac{1}{2}} \frac{\xi^4}{(1-4\xi)(1-\xi^5)} \, d\xi$$

$$= 2\pi i \operatorname{Res}\left[\frac{\frac{1}{4}\xi^4}{(\xi-\frac{1}{4})(\xi^5-1)}, \frac{1}{4}\right]$$

$$= 2\pi i \frac{\frac{1}{4}\cdot\frac{1}{4^4}}{\frac{1}{4^5}-1} = 2\pi i \frac{1}{1-4^5} = -\frac{2\pi i}{1023}$$

11. $$\int_{-\infty}^{+\infty} \frac{(\cos 3x)^2}{x^2+4x+5} \, dx = 2\pi i \operatorname{Res}[f(z), -2+i]$$

$$= 2\pi i \left.\frac{(\cos 3x)^2}{x+2+i}\right|_{x=-2+i}$$

$x^2+4x+5=0$

$(x+2)^2=-1$

$x_1=-2+i$ $\quad x_2=-2-i$

$$\frac{(\cos 3x)^2}{[x-(-2+i)][x-(-2-i)]}$$

上半平面的奇点

$$(\cos 3x)^2 = \frac{\cos 6x+1}{2}$$

$$I = \int_{-\infty}^{+\infty} f(x) \, dx$$

当 $|z|\to\infty$ 有 $|zf(z)|\to 0$